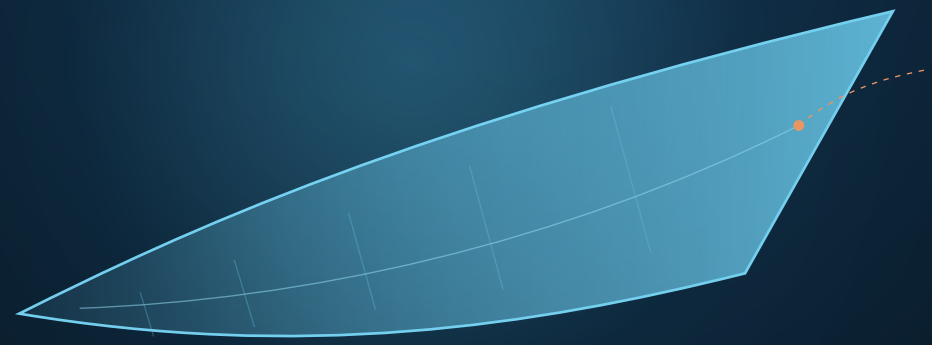
STYLIANOS APOSTOLOU

AEROSPACE ENGINEER

Propulsion · Multidisciplinary Modelling · Advanced Aircraft Systems

Hybrid-Electric Propulsion | Aeroelasticity | Thermal Management | Cryogenic Structures



IMPERIAL COLLEGE LONDON

MSc Advanced Aeronautical Engineering · 2026-2027

UNIVERSITY OF PATRAS

Diploma in Aeronautical Engineering · Final grade 8.47/10

ENGINEERING PROFILE

Physics-based modelling across propulsion, flow and structures.

A multidisciplinary aerospace profile built around turning coupled physical systems into computational models, design trade-offs and engineering decisions.

PROFESSIONAL SUMMARY

Aerospace Engineer currently pursuing an MSc in Advanced Aeronautical Engineering at Imperial College London, with project experience spanning hybrid-electric propulsion, aircraft thermal management, structural dynamics, aeroelasticity and lightweight structural design. My work focuses on physics-based computational modelling to investigate multidisciplinary aerospace systems, quantify design trade-offs and translate simulation results into engineering decisions.

MATLAB · Simulink · Python · Numerical Methods · System Modelling

01

PROPULSION & ENERGY

Hybrid-electric propulsion · aircraft thermal management · electric propulsion · cryogenic & hydrogen systems.

02

MODELLING & SIMULATION

MATLAB/Simulink · dynamic systems · numerical integration · state-space methods · parametric analysis.

03

AEROSPACE ANALYSIS

Advanced aerodynamics · structural dynamics · aeroelasticity · heat transfer · fluid mechanics.

04

ENGINEERING DESIGN

System-level trade studies · lightweight sizing · mission-based analysis · technical literature synthesis.

NOW
Imperial College London

FOCUS
Propulsion · Aerodynamics · Modelling

NEXT
Advanced aerospace R&D

Integrated Power & Thermal Management for Hybrid-Electric Aircraft

Mission-level modelling and evaluation of advanced cooling architectures for an ATR-72-class hybrid-electric propulsion system · Diploma Thesis · 2026

ENGINEERING CHALLENGE

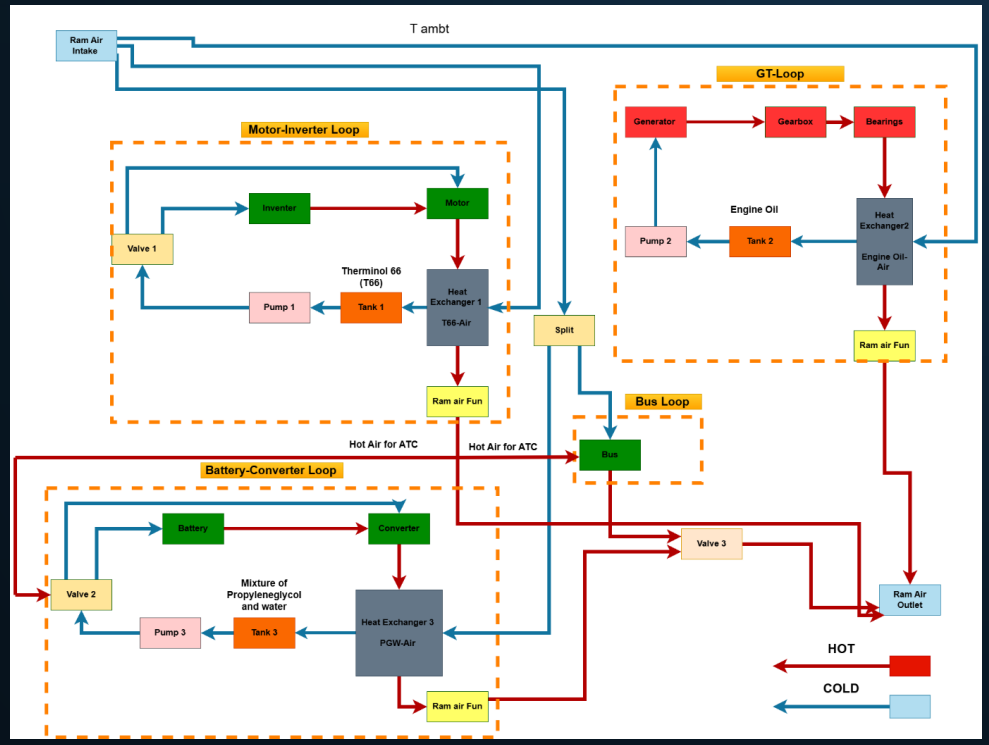
Electrified aircraft propulsion introduces high, transient and distributed thermal loads across electric machines, power electronics and energy storage. The objective was to maintain component temperatures within operating limits while evaluating hydraulic, energetic and integration penalties across alternative cooling strategies.

MODELLING APPROACH

A coupled MATLAB/Simulink framework converted mission-dependent electrical losses into component heat loads and thermal response. Cooling-loop operation was evaluated through coolant mass flow, pressure losses and pump-power requirements across time-varying aircraft operating conditions.

DESIGN OBJECTIVE

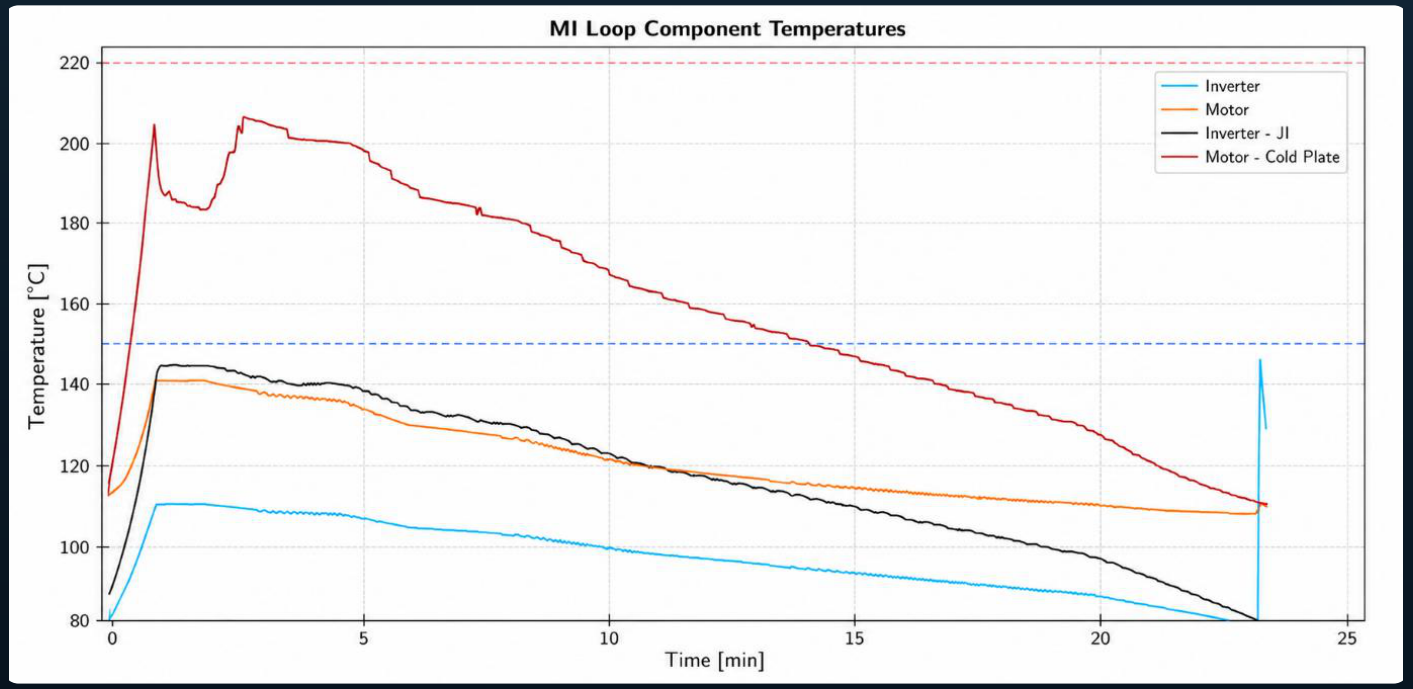
Identify the best **system-level balance** between thermal protection, hydraulic demand, energy consumption and integration penalties - not simply the strongest local cooling technology.



MISSION PROFILE | POWER DEMAND | LOSSES / HEAT | THERMAL RESPONSE | COOLING REQUIREMENT | FLOW + ΔP | PUMP ENERGY

From local cooling performance to system-level optimisation.

Baseline cold plates → PCM-assisted buffering → PCM + targeted jet impingement.



−71.35%

TOTAL PUMPING ENERGY HYBRID VS BASELINE

−54.61%

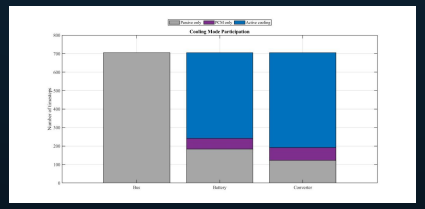
MEAN MI-LOOP COOLANT FLOW 8.02 → 3.64 KG/S

−39.52%

BC-LOOP PUMPING ENERGY WITH PCM

148.1 → 145.1°C

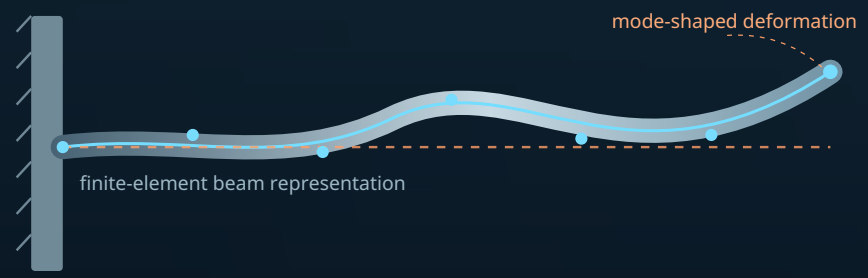
PEAK INVERTER TEMPERATURE WITH JET IMPINGEMENT



ENGINEERING INSIGHT — No single cooling technology was optimal across all components and operating conditions. PCM was most effective as a transient thermal buffer, while jet impingement targeted the inverter's high local heat flux. The strongest architecture emerged from targeted technology allocation at system level; local improvements could also redistribute thermal and hydraulic penalties elsewhere in the aircraft TMS.

Dynamic & Aeroelastic Analysis of an Aircraft Wing

Finite-element modal analysis · forced vibration · damping · time-domain aeroelastic simulation in MATLAB · Two-person team project



3.22 Hz	20.18 Hz	25.88 Hz	56.41 Hz	77.64 Hz
Mode 1	Mode 2	Mode 3	Mode 4	Mode 5

19.99 Hz engine excitation ≈ 20.18 Hz second mode



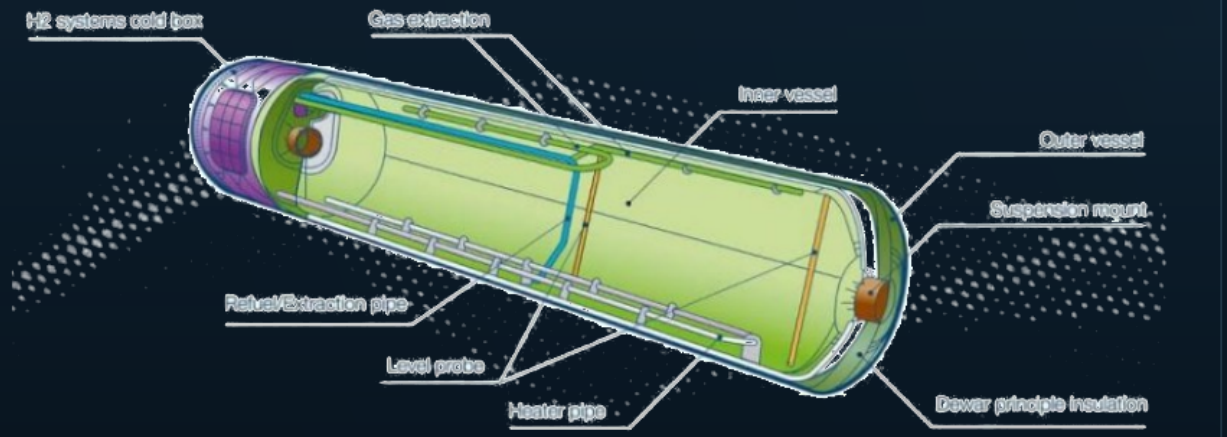
Approach: 120 finite elements · 363×363 global mass/stiffness matrices · 3 DOF/node · modal superposition · stiffness-proportional damping · central difference and Newmark integration.

ENGINEERING INSIGHT

Near-resonant forcing amplified the second structural mode. In the aeroelastic extension, the more flexible I-beam representation developed static divergence at 70 m/s, showing how structural idealisation and torsional stiffness alter predicted stability.

Lightweight Structural Design of a Cryogenic LH_2 Pressure Vessel

Thin-walled structural analysis and parametric sizing under combined pressure, inertial and thermal loading · Two-person team project



PRESSURE | INERTIAL | THERMAL

SELECTED LIGHTWEIGHT CONFIGURATION

7.849 kg

Candidate geometries were screened using equivalent stress and safety-factor constraints before ranking feasible configurations by structural performance and mass.

1.00 mm Vessel wall thickness	**2.45 mm** Stringer thickness
45 / 15 / 35 mm Height / foot / head	**7 layers** CFRP stringer laminate
1.623 Critical factor of safety	**≈1.25%** Deviation from FoS target

GEOMETRY | PRESSURE / INERTIAL / THERMAL | N-Q-M | SECTION PROPERTIES | STRESS + SHEAR FLOW | VON MISES / FoS | MASS / SELECTION

ENGINEERING INSIGHT — Thermal loading dominated the normal-stress state, while shear stresses remained comparatively small. Parametric sizing transformed the analytical stress model into a practical design tool for selecting a lightweight configuration close to the prescribed safety-factor target.

Thermal Management in More-Electric & Hybrid-Electric Aircraft

Technology assessment and system-level synthesis of thermal-management architectures for aircraft electrification.

01

DISTRIBUTED HEAT SOURCES

Electric motors · power electronics · batteries · generators · avionics · propulsion systems.

02

THERMAL TECHNOLOGIES

Cold plates · PCM · jet impingement · spray cooling · heat pipes · compact heat exchangers · cryogenic heat sinks.

03

INTEGRATED TMS

Propulsion · electrical system · fuel system · environmental control · thermal-management coordination.

04

AIRCRAFT-LEVEL TRADE-OFFS

Temperature · mass · energy · volume · reliability · complexity · mission dependence.

LITERATURE SYNTHESIS → technology landscape & system challenges → **ENGINEERING QUESTION** → how should cooling technologies interact over a mission? → **DIPLOMA THESIS**

RESEARCH OUTCOME — Aircraft electrification shifts thermal management from isolated, predominantly steady heat sources toward distributed and mission-dependent thermal loads. This study established the technical foundation for the subsequent diploma thesis, moving from technology review to integrated power-and-thermal system modelling.

CURRENT ENGINEERING WORK

Advanced Aeronautical Engineering at Imperial College London

Current MSc study — presented as technical development, not completed project experience.

CURRENT MSc STUDY

Turbulence & Turbulence Modelling

Turbulent-flow physics · statistical description · modelling approaches · engineering interpretation.

CURRENT MSc STUDY

Applied Aerodynamics

Advanced aerodynamic analysis · viscous-flow behaviour · aircraft aerodynamic applications · physics-based flow interpretation.

CURRENT MSc STUDY

Fundamentals of Scientific Machine Learning

Scientific computing · data-driven modelling · machine-learning methods for physical systems.

CURRENT MSc STUDY

Emerging Technology for Green Aviation

Aircraft decarbonisation · emerging propulsion technologies · energy-system integration · sustainable aviation concepts.

CURRENT RESEARCH FOCUS

Electric Propulsion & Plasma Diagnostics

Preparatory research into plasma-thruster plume characterisation, charged-particle motion, E×B velocity-filter diagnostics, ion velocity / charge-state populations, vacuum-facility testing and experimental interpretation.

ENGINEERING TOOLKIT

Tools, methods and technical foundation.

MODELLING & SIMULATION

MATLAB · Simulink · dynamic system modelling · numerical integration · state-space analysis · parametric optimisation.

AEROSPACE ENGINEERING

Aerodynamics · fluid mechanics · heat transfer · thermal management · structural dynamics · aeroelasticity · stress analysis.

PROGRAMMING

MATLAB — advanced academic/project use
Simulink — system modelling
Python — engineering/scientific computing
C — programming fundamentals
C++ — introductory

ENGINEERING METHODS

Finite-element modelling · modal analysis · parametric design · mission-based simulation · system-level trade studies · technical literature synthesis.

ENGINEERING SOFTWARE

AutoCAD · Autodesk Fusion 360 · MATLAB/Simulink environment · scientific Python workflows.

EDUCATION

Imperial College London — MSc Advanced Aeronautical Engineering, 2026-2027.
University of Patras — Diploma in Aeronautical Engineering (5-year Integrated Master's level), 2021-2026 · 8.47/10.

CONTACT

STYLIANOS APOSTOLOU

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London, United Kingdom

PROPULSION · MODELLING · MULTIDISCIPLINARY AEROSPACE SYSTEMS

Portfolio prepared for engineering and aerospace R&D applications.